Bounding Attribution Sensitivity in Feed-Dependent Airline Route Decisions: A Three-Case Study with a Scalable Screening Approach

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July 2026

Abstract

Airline route go/no-go decisions on hub-fed markets require attributing connecting-passenger revenue across the segments that carried the passenger. Two attribution rules are commonly considered: mileage proration (industry standard) and Shapley value (from cooperative game theory). This work asks whether the choice of attribution rule can move the profit calculation enough to flip the verdict on real US spoke-to-hub markets. DESTER implements a three-verdict engine — local viability, feed-added viability, and attribution sensitivity — grounded in Bureau of Transportation Statistics ticket-level data and carrier-level cost data. A multinomial logit share model is calibrated on 219 comparable domestic markets. The initial fit reveals a systematic transferability failure driven by ultra-low-cost carrier fixed-effect intercepts, resolved via empirical-Bayes shrinkage on carrier intercepts (backtest MAE 9.94 percentage points, from 13.34pp unregularized). Three rigorously analyzed markets span an empirical leverage bracket: AUS-SLC (13

Introduction

An airline evaluating a proposed route must decide whether the anticipated revenue justifies the operating cost. On markets where a substantial share of passengers connect through a hub — behind-and-beyond traffic — a further question arises: how should the connecting fare be attributed across the segments the passenger flew? A single itinerary AUS→SLC→SEA is one ticket, one fare, and two flights. The AUS-SLC leg and the SLC-SEA leg both contributed to earning that fare.

Two attribution rules are common. *Mileage proration* allocates fare in proportion to distance: $\phi_A = \text{fare} \cdot (m_A/m_{\text{total}})$. This is the historical interline settlement default and remains the operational baseline in many airline systems. *Shapley value*, from cooperative game theory, allocates by expected marginal contribution to the coalition:

$$\phi_A = \tfrac{1}{2}\big[v(A) + (v(A,B) - v(B))\big]$$

where $v(A)$ is the fare AUS-SLC would earn as a standalone product, $v(B)$ is SLC-SEA's standalone fare, and $v(A,B)$ is the connecting fare actually paid. By construction, $\phi_A + \phi_B = v(A,B)$: Shapley splits the fare exactly, but weights the split by each segment's standalone earning power rather than by geographic length.

The two rules can disagree substantially. On a market where the trunk segment is a large fraction of total mileage but the feed segment has a disproportionately high standalone fare per mile — for example, a short regional segment in a monopoly market — mileage proration will over-credit the trunk while Shapley will credit the feed more generously. DESTER asks: on real US spoke-to-hub markets, is this disagreement large enough to change the airline's go/no-go decision?

The research is structured around a pre-registered running market (AUS-SLC on Delta, chosen for its structural similarity to a recently-announced network expansion), extended after initial results to a second market (ATL-SAT, drawn from a systematic screener) and then a third (MIA-SEA, drawn from a leverage-focused screen). A scalable variant of the pipeline computes attribution leverage across 921 additional US spoke-to-hub markets, producing the first empirical characterization of Shapley-vs-mileage sensitivity at population scale.

Method

Data sources

All three data sources are public, from the US Bureau of Transportation Statistics.

- *DB1BMarket 2025 Q2* — 10% sample of ticketed domestic itineraries, approximately 8.5 million rows, aggregated at the (origin, destination, carrier, coupon-count) level.
- *Form 41 Schedule P-5.2* — quarterly aircraft operating expenses by carrier and aircraft type.
- *T-100 Segment* — monthly aircraft operating statistics by carrier, aircraft type, and O&D segment.

DB1B, historically the O&D survey collected quarterly at a 10% sample rate, was discontinued in July 2025 and replaced with DB1C (monthly, 40% sample). This work uses the final DB1B quarter (2025 Q2).

The three verdicts

Verdict 1 — Local viability. Predicts point-to-point O&D demand on the market. Steps: sample market size from DB1B; annualize; apply stimulation uplift for induced demand; predict operating carrier's share via a multinomial logit; apply the airline-industry S-curve for the frequency-share effect; model daily demand as Poisson around the predicted mean, cap at seat capacity, apply a recapture rate on spilled passengers; compute revenue and cost; verdict from whether contribution is non-negative.

Verdict 2 — Feed contribution. Adds behind-and-beyond feed passengers routed through the hub end of the O&D. Filter DB1B to two-coupon itineraries where one leg is the trunk market and the itinerary market pair itself is not the trunk market. Compute observed share of each feed-market by the operating carrier. Recompute contribution with combined local + feed revenue against unchanged cost.

Verdict 3 — Attribution sensitivity. For every feed itinerary, compute the trunk segment's fare allocation under both regimes: - Mileage: $\phi_A = \text{fare} \cdot (m_A/m_{\text{total}})$ - Shapley: $\phi_A = 0.5 \cdot (v(A) + (\text{fare} - v(B)))$

Total contribution is then recomputed under each regime, and the verdict flip is reported if any of 48 pre-registered (uplift, recapture, α) combinations produces disagreeing verdicts between regimes.

The share model and its transferability failure

The multinomial logit share model uses six features per (carrier, itinerary-type) alternative: `log_fare`, `n_stops`, `routing_efficiency`, `frequency_weekly`, `hub_dominance` (a binary indicator for whether the operating carrier holds $\geq 40\%$ share at the relevant hub), and carrier fixed-effect intercepts. Coefficients are estimated by maximum likelihood using statsmodels' MNLogit implementation with L-BFGS optimization, after standardizing features to prevent Newton-step divergence.

Backtest on 30% held-out markets from the calibration set produced mean absolute error of 13.34 percentage points on carrier-level share — within the range acceptable for published airline logit models. However, application of the fitted model to a second market (ATL-SAT), unrefit, produced a predicted Delta share of 0.9% against an observed share of 58.5%, a 64× miscalibration.

Diagnosis. Ultra-low-cost carriers Frontier (F9) and Spirit (NK) had large positive fixed-effect intercepts ($\beta_{F9} = +6.77$, $\beta_{NK} = +6.01$) fit to a subset of markets where they held unusually high share. Their inflated intercepts dominated the softmax on the second market. The problem was neither absence (both carriers appeared in 800+ training rows) nor market-diversity thinness (both appeared in 94+ distinct training markets). It was within-market share concentration.

Empirical-Bayes shrinkage on carrier intercepts

Two threshold-based fixes failed cleanly: minimum row-count filtering and minimum-distinct-markets filtering. The fix that worked applies post-fit shrinkage:

$$\beta_c^{\text{new}} = \beta_c^{\text{old}} \cdot \frac{1}{1 + \lambda \cdot \sigma_c^2}$$

where σ_c^2 is the variance of carrier c's per-market share-prediction residual across the training set. High-variance intercepts are shrunk toward zero; low-variance intercepts pass through nearly unchanged. At $\lambda = 15$ (chosen from two candidate values via the lower-λ tiebreak rule), F9 intercept shrinks by 61%, NK by 49%, both clearing the pre-registered 40% threshold. Backtest MAE improves to 9.94pp, clearing the 10pp publishable-quality target.

Applied to the second market, Delta's predicted share rises from 0.9% to 23.8% — a 26× improvement, though still below observed 58.5%. The residual gap is attributable to systemic aggregation effects and is documented, not resolved.

Findings

Three rigorously analyzed markets

Market	Feed share	Leverage	Verdict flipped?
AUS-SLC (Delta)	13%	5.74%	No
ATL-SAT (Delta)	54%	33.3%	No
MIA-SEA (Delta)	9%	1.3%	No

The attribution mechanism has material leverage — up to $5.4M annually on ATL-SAT — but does not flip verdicts on any of the three markets, all robust across 96 pre-registered sensitivity combinations.

The AUS-SLC vs. ATL-SAT comparison established the primary empirical bracket: 5.74% to 33.3% attribution leverage across two structurally comparable markets separated by feed share (13% vs. 54%). The MIA-SEA result contradicts the initial hypothesis that feed share monotonically predicts leverage: MIA-SEA has feed share (9%) lower than AUS-SLC, yet lower leverage than both other markets. Feed share is a necessary but not sufficient condition for attribution leverage.

Mechanism finding: direction of the Shapley-vs-mileage delta

On AUS-SLC, Shapley credits the AUS-SLC segment *less* than mileage does, by an average of $79 per feed itinerary. Explanation: short regional segments from SLC (SLC-Boise, SLC-Jackson Hole, SLC-Kalispell) carry disproportionately high fares per mile — regional monopoly pricing. Distance-based mileage proration over-credits the AUS-SLC trunk relative to what its standalone value would earn. Shapley's value-weighted split corrects this.

The direction of the Shapley-vs-mileage delta on any trunk segment depends on the relative per-mile yields of the trunk and its feeders. On markets with the opposite yield structure — long-haul high-yield trunk with regional feeders in low-yield markets — Shapley would credit the trunk more, not less.

Population-scale screening result

A scalable variant of the pipeline was applied to 921 US spoke-to-hub domestic markets. This variant uses aggregated features (ratio-of-sums approximations) and evaluates the full 96-point sensitivity grid per market.

- 106 markets (11.5%) contained verdict flips somewhere in their pre-registered sensitivity space.
- The subpopulation of flipping markets skewed smaller in local passenger volume (median 1,400 sample vs. 1,827 for non-flipping).
- The single market meeting the joint criteria of AUS-SLC-comparable size, verdict-flipping, and Delta dominant carrier was MIA-SEA. This was chosen as the third rigorous case study.

Rigorous re-analysis of MIA-SEA did not reproduce the fast-screener's flip prediction. The screener estimated 36% attribution leverage; the rigorous pipeline computed 1.3%. This is not a bug — it is the honest correspondence between fast approximation and full analysis, and it establishes the effective error bar on the population-scale screening result. The 106-flip finding represents an upper bound of potentially-flipping markets.

Limitations

Aggregate feature approximation. The on-demand and catalog engines use ratio-of-sums approximations of the ratio-of-ratios features the rigorous pipeline computes exactly. This is a documented and inherent cost of the aggregation step. MIA-SEA's non-reproduction of the screener flip prediction is the quantified illustration of this limitation's magnitude.

CASM basis. Delta mainline does not operate the E175 used on AUS-SLC; SkyWest operates it under a capacity-purchase agreement. DESTER uses SkyWest's operating cost plus a 12% CPA markup — an industry-typical central estimate.

Standalone-fare sparsity. For thin-spoke markets, up to 48% of feed itineraries have no standalone fare data because the endpoint city lacks nonstop service to connecting destinations. Shapley collapses to mileage-equivalent for these, structurally bounding the mechanism's leverage on thin-spoke markets.

Population screening is upper-bound. The 106-flip screening result is an upper bound; the true number of markets with rigorous flips is likely lower.

Discussion

DESTER's central research question was whether the choice between mileage proration and Shapley value can flip the go/no-go verdict on real US airline route decisions. On three rigorously analyzed markets spanning the full range of feed share from 9% to 54%, the answer is no. However:

- The mechanism produces material dollar swings on markets with substantial feed. On ATL-SAT, the choice of attribution rule moves the annual contribution figure by $5.4M. This is not a rounding error at the scale of joint-venture settlements or code-share pricing negotiations.
- The mechanism's direction is not universal.
- Feed share is not a sufficient predictor of leverage. MIA-SEA (9% feed) has lower leverage than AUS-SLC (13% feed).
- Verdict flips exist in principle. The population-scale screener flags 11.5% of US spoke-to-hub markets as containing flips in their operational parameter space. The single most promising rigorous candidate did not reproduce a flip.

The methodological contribution is arguably as valuable as the substantive finding. The share model's transferability failure required a shrinkage-based fix that generalizes well: any airline logit model calibrated on a heterogeneous market population is likely to encounter similar within-carrier share-concentration effects on ULCCs or leisure-focused carriers, and post-fit shrinkage on high-variance intercepts is a lightweight remedy that does not require re-estimation.

Conclusion

Feed-dependent airline route decisions are the kind of decision an airline network planner makes routinely: is a new nonstop justified by local demand alone, does adding hub feed make the case stronger, and does the attribution rule for connecting fare change the answer? DESTER answers all three for three rigorously analyzed markets and characterizes the attribution mechanism's behavior across an additional 921 markets at population scale.

Attribution leverage exists and matters in dollar terms. Verdict flips are possible in principle and empirically flagged in a non-trivial minority of markets. In the three cases rigorously tested, verdicts held under both regimes. The story is more nuanced than the initial hypothesis anticipated, and — via the honest documentation of failed intermediate iterations — DESTER's methodology traces a path a real research project follows when the data pushes back.

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